

Executive Summary

Recent ADHD research underscores *externalising cognitive tasks* to compensate for internal deficits. Meta-analyses show large, consistent time-perception impairments in ADHD: children/adolescents with ADHD “perceived time less accurately (Hedges’ $g > 0.40$) and less precisely ($g \approx 0.66$)” than controls [16†L52-L60]. ADHD also shows steeper delay-discounting: a stronger preference for small immediate rewards over larger delayed ones [31†L127-L135]. Similarly, everyday prospective memory is often impaired, so external cues (event-based triggers) can restore performance [36†L426-L435]. Mind-wandering is a core ADHD symptom [41†L50-L59], meaning users will drift unless the device provides re-engagement cues without blame.

The device design must therefore *reduce executive load*: supply defaults, cues, immediate feedback, and “lazy paths” to action. Key micro-interventions include: displaying **remaining time** (to offload timing) while avoiding attention-grabbing animation; using a **rotary dial + one-press start** to minimise choices; giving *immediate, small rewards* (points) at task-start and completion; employing **NFC context tags** as external cues; automatically **resuming** unfinished tasks; and suggesting tiny **movement breaks**. Each element is grounded in literature (see Table below and citations).

This report synthesises primary sources on ADHD timing, attention, motivation and cueing to recommend exactly how each UI choice should be implemented. For each micro-decision we state the evidence, design proposal, possible failure modes, and hypotheses to test. We then outline an **experimental protocol** (RCT with ADHD teens) for the top hypotheses. Finally we deliver detailed UI specs: a front-panel layout, five screen-state mockup descriptions (Idle, Active, Claim-ready, NFC checkpoint, Night), pixel/font/color guidelines, and mermaid diagrams (dial state machine, session timeline, NFC flow).

Key findings: (1) Always show “time remaining” (countdown) as the hero display [16†L52-L60], supplementing it with a static progress bar or small elapsed indicator. Avoid every-second movement on the screen except the timer itself. (2) Use one accent colour (e.g. hex #FF6633) exclusively for “active/attention” elements (time remaining, buttons in action states), and neutral white/grey for all else, so attention is driven by state semantics. (3) The device must *presume distraction* will happen: never punish pausing or abandoning. Instead, *persist state* so users can **resume** where they left off without guilt. (4) Reward *task initiation* with an immediate +1 point, plus additional points on completion [31†L127-L135]. Make confirming completion a deliberate action (CONFIRM button) to create a satisfying ritual. (5) Use NFC tags purely as “when X happens, prompt Y” triggers [36†L426-L435]: e.g. “At home → Show next task” (see mermaid flowchart below). (6) Encourage movement during breaks: short “stretch/walk” suggestions can improve focus on return [50†L285-L293]. (7) Avoid complex gamification: small non-negative points (never deductions) and descriptive progress (“3 sessions today”) are better than points tallies dominating the screen. (8) Defaults: preselect the next task and a conservative session length (e.g. 20 min) so users just press START. Make cancelling tasks slightly harder (press+hold) but starting/pausing easy. (9) Testable hypotheses include: *Countdown vs count-up* effects on completion; *progress bar vs none*; *default start vs manual select*; *NFC cueing vs no cue*; *points vs none*; *stimulus pulse vs none*; etc. We propose an 8-hypothesis within-subject trial with ~30 ADHD teens to measure start-rate, on-task time, return latency and user preference (see Experimental section).

All recommendations emphasise **low distraction**: nearly static displays and minimal animation. The screen should become *boring during focus* to avoid competing for attention. Overall, the device is framed as an “*external executive-function instrument*”, shouldering memory and timing tasks so the student doesn’t have to.

1. ADHD and Time Perception

Adults and children with ADHD consistently show impaired temporal processing. Meta-analyses found *large deficits* in time discrimination, estimation, production and reproduction tasks [10†L21-L30]

[10†L41-L50] . Zheng et al. (2022) meta-analysed 27 studies (1,620 ADHD vs 1,249 controls) and reported that ADHD youth were **less accurate (Hedges’ $g > 0.40$) and less precise ($g = 0.66$)** in perceiving time, tending to **overestimate** durations [16†L52-L60] . This aligns with an “accelerated internal clock” model: ADHD individuals often lose track of passing time.

Design implication: Show time externally. In practice, this means a prominent countdown or time display on the main screen. Giving “time remaining” makes the *end point concrete*, countering ADHD time-blindness [13†L89-L97] [16†L52-L60] . It can help with delay aversion (ADHD children prefer shorter waits [31†L127-L135]). However, the interface should not require complex calculation: a simple numeric readout and perhaps a progress bar (see Section 2). If we lacked any evidence, at minimum we know “seeing time pass” helps ADHD, so a visible timer is justified [13†L89-L97] [16†L52-L60] .

2. Reward Sensitivity & Reinforcement

ADHD is linked to altered reward processing. Multiple meta-analyses show ADHD individuals disproportionately prefer **immediate rewards** over larger delayed ones [31†L127-L135] . This “delay aversion” means that any reward must be closely tied to the behavior. Accordingly:

- **Immediate reinforcement:** The device should reward the user *as soon as possible*. A simple +1 point for *starting* a task session (instead of only at the end) provides instant positive feedback [31†L127-L135] . Then additional points for midpoints (5 min) or completion give quick payoffs. This leverages ADHD’s strong response to reinforcement (including both reward and avoidance of loss) [31†L127-L135] .
- **Small, frequent rewards:** Rather than “scoreboard” style, show points in small increments and mostly for achievements. For example, “+3” flashes when confirming task completion.
- **No penalties:** Avoid negative feedback. Research suggests punishment can exacerbate ADHD symptoms (avoidance, demotivation). Instead of a “fail” state or point loss on interruption, use neutral states (“PAUSED”) and persistent resumability (below).
- **Descriptive stats:** Present progress as factual readouts (“Today: 47 min total”) rather than cheering messages. Descriptive feedback is less likely to induce shame or stress.

Evidence: A systematic review found that operant reinforcement (points, tokens) reliably improves ADHD task performance [31†L127-L135] . While gamified systems have mixed evidence, reinforcement has the strongest support. Thus, **points/feedback are justified**, but should be implemented carefully (see Micro-Decision: Gamification).

3. Prospective Memory & NFC Cues

Prospective memory (remembering to do something later) is often impaired in ADHD. However, external cues can dramatically improve it. **Event-based cues** (triggered by an external event) require much less

cognitive monitoring than pure time-based tasks [36†L426-L435] . NFC tags serve exactly this role: “When I tap this tag, do X.” In practice:

- **No password logic:** The NFC tags are not ID or context authentication per se, but simple triggers. E.g. “Arrived at home” tag → switch device to homework mode (show next task). “Morning” tag → list today’s tasks. “Desk” tag → open focus screen.
- **Implementation intention:** The device effectively encodes “If [context], then [action]”. This ties into research on implementation intentions, which in adults yields a **medium-large PM improvement** ($d \approx 0.6$) [36†L399-L407] . Children are more variable, but linking a physical action (tapping) to a digital response should reduce forgetting.
- **Prospective vs retrospective:** The device’s UI should clearly show *what the intended action is*. For example, after an NFC scan, display “HOME – Next: MATHS 20:00”. This clarifies both the cue and the response.

Evidence: A meta-analytic review showed implementation intentions help executive control, though effects are sometimes smaller in children [36†L399-L407] . Nevertheless, using physical NFC is likely more effective than a mental plan. We should also ensure the user associates the tag locations (if physical tags) or triggers with tasks in advance.

4. Mind-Wandering and Focus

Mind-wandering (“task-unrelated thought”) is dramatically elevated in ADHD [41†L50-L59] . Indeed, DSM-5 symptoms “mind seems elsewhere” acknowledge this. Thus:

- **Expect disengagement:** The device should assume the user will repeatedly lose attention. It must be *non-judgemental* (no “FAILED” screens).
- **Re-engagement cues:** If a session continues ticking after distraction, the interface can subtly re-prompt upon return. For example, when the user returns and unlocks the device, show the remaining time and task name immediately (as “RESUME”).
- **Silent persistence:** Rather than audio alarms, a very brief visual pulse (e.g. the countdown digits flicker or the border flashes once) could alert the user to refocus without a prolonged interruption. This could be on a timer (e.g. every 10 min) or triggered by the system sensing inactivity (if hardware permits). The key is *brevity and subtlety* – avoid loud beeps or full-screen animations that break concentration aggressively.

Evidence: Mind-wandering correlates with ADHD core symptoms in adults [41†L50-L59] . Cognitive models (e.g. executive failure hypothesis) suggest ADHD brains need help suppressing wandering. Our design leverages “stimulus-driven attention” (e.g. a flash) to snap back focus, without penalising the user [41†L50-L59] .

5. Countdown vs Count-Up vs Hybrid (Micro-Decision)

There is no definitive evidence that countdown is better than elapsed time for ADHD. Both have psychological effects:

- **Countdown (“time remaining”):** Emphasises a *finite deadline*. May motivate by showing the end point approaching. However, it *constantly reminds* the user of time leaving, which can induce anxiety or cause the user to watch the clock. ADHDers often dislike ticking deadlines, but many coaching sites still recommend visible countdowns (e.g. Time Timer).

- **Count-up (“time elapsed”):** Highlights progress (“I’m 5 minutes in”). Could create a “now I’ve started” momentum. It never shows how much is left, so it may reduce pressure but might cause confusion about session length. The user must remember (or trust) the planned duration.
- **Hybrid (both):** Show **both** remaining and elapsed. This adds cognitive load (two numbers) but could give full information: “14:32 left +05:28 done”. Alternatively, a progress bar or subtle indicator (minutes done) along with the main countdown can offer a quick readout of progress.

Design recommendation: Use a **countdown timer** as the default, because externalising the **remaining time** directly addresses ADHD timing deficits 【16†L52-L60】. Supplement it with a small static progress bar (see Section 6) or a tiny “elapsed” text (e.g. “+05:28”) if space permits. The progress bar provides an at-a-glance sense of magnitude, mitigating the single-number limitation. Optionally, let the user *toggle* display mode with the dial (e.g. press-on dial switches between “TIME REMAINING” and “TIME ELAPSED”).

Potential failure: Countdown could make users anxious or hyper-focus on time. The progress bar mitigates this by being a quiet visual cue. This is a hypothesis to test: e.g. A/B test countdown-only vs count-up-only vs both on task persistence.

See Table below for a detailed countdown vs count-up vs hybrid comparison.

6. Granularity: Seconds vs Minutes; Numeric + Spatial Encoding

Seconds vs whole minutes: Counting seconds on-screen means the display changes every second. This can be visually stimulating (movement of digits) and might distract. A compromise: show minutes and seconds, but update seconds by “jumping” every second, while keeping surrounding text static. Or only show minutes and a “:00” that changes occasionally. For example:

14 : 32

The large numbers tick, the bar or other UI stays fixed.

Bar chart / progress indicator: Adding a horizontal bar (▀ style or filled rectangle) under the timer conveys the ratio done/remaining. This is often called a “progress bar.” Research in related domains (e.g. waiting times) suggests graphical countdowns can shorten perceived wait 【45†L1-L4】, likely by engaging visual estimation. Here, a bar won’t “flash,” it simply grows or shrinks slowly with time. This dual encoding (numeric + bar) gives ADHD users two ways to sense time without conscious calculation.

Design: We recommend a thick single-segment bar under the time, filling from left (start) to right (end). For example:

14 : 32

(Here the bar shows 70% filled.) The bar color is the same orange accent. This bar should update perhaps every minute or less frequently, not every second, to avoid constant visual change.

Trade-off: A bar provides intuitive magnitude but needs careful tuning. It shouldn't animate every second (would be a slow movement or discrete jumps). We can update it, say, every 10 seconds or minute. The main timer number is primary.

7. Animation and Motion

To keep the UI **low-distraction**, animations should be minimal. The countdown digits will change, but other elements must remain static. We adopt the principle: **move only what must move**.

- **Good motion:** A brief (0.2–0.3 s) flash or pulse on the timer/confirm button when a key event occurs (session start, end, or interrupt). One could quickly tint the timer orange on start, then fade to grey, for example. The pulse should be localized.
- **Bad motion:** Continuous animation (rotating graphics, bouncing icons, moving particles) would break concentration. Even moving progress circles or scanning effects are probably too engaging. Avoid any background changes.
- **Beep/haptic:** A single “click” or short beep when the session starts or ends can help (if not annoying). Use a simple tone (e.g. 800 Hz, 0.1 s) only at start/stop. A mechanical click from the dial is nice for the user (they'll hear their own action).

Rationale: HCI studies show flashy animations can attract attention and delay resumption of the primary task. Since the aim here is *boring focus*, animations are cues, not entertainment. All dynamic feedback should be as brief as possible.

8. Color, Contrast, Typography

- **Colors:** Use a **near-black** background (e.g. #121212). Main text/icons in white (#FFFFFF) or light grey (#DDDDDD). A single orange/coral accent (e.g. #FF6633) is used *only* for active elements: the numeric timer, progress bar, Start/Pause/Confirm buttons when enabled. Inactive or context text uses grey. This ensures “orange = needs attention”. For example, when counting down, the digits and bar are orange; when paused or after finish, they could turn grey.
- **Contrast:** Maintain WCAG AA contrast: orange text on black is fine (for 120px digits), white on black is obvious, grey on black is subtle but legible.
- **Fonts:** Prefer a utilitarian sans-serif (like DIN or a thin Grotesk) similar to Teenage Engineering style. All text in sentence case. Suggested sizes (for 240×320px active area):
 - Hero timer digits: ~100–120 px (digit height ~14 mm).
 - Task name: ~24–28 px (e.g. “CHEMISTRY”).
 - Status labels (“NEXT”, “TODAY”, “COMPLETED”): ~14–18 px, uppercase or small-caps.
 - Supporting text (“Atoms · Q1–8”, “24 pts”): ~12–14 px.
- Footer/datetime: ~10 px (tiny).
- **Micro-labels:** As per TE aesthetic, small printed labels under icons or at corners can add character (e.g. “TODAY” above date, or a tiny “task engine · NFC · wake-word” under the wordmark). Keep these extremely small (~8px) and unintrusive.

9. Physical Interaction & Friction

The dial and buttons should create an interaction hierarchy: easy to do the *right* thing, slightly harder to do the *wrong* thing.

- **Dial (task selector + time):** Make the dial the primary control. Rotating it steps through tasks or time presets. The dial gives continuous tactile feedback (clicky knurl, 6 mm shaft with set-screw). Movement is easy for starting and harder to hide from.
- **Start/Pause buttons:** Two 12–15 mm buttons next to the dial. Press to start or pause. They should be very easy (light click).
- **Press vs press+hold:** **Start/Pause** are single-click. For *undesirable actions* (e.g. cancel task or reset session), require press-and-hold (~1 s). This reduces accidental cancellations.
- **LED cluster:** An indicator cluster could show Bluetooth/Wi-Fi or recording status. Keep small (3–5 mm LED holes) and static icons.
- **Switches:** Slide switches for mic mute and main power (13×6 mm). Label them tiny.

This asymmetric friction ensures that pressing Start (the “lazy route”) is frictionless, while task deletion or resetting requires extra intent (hold+turn).

10. Confirmation Ritual

When a session ends, require the user to *confirm* completion explicitly. This has two benefits: it provides a natural breakpoint for reward (points, words), and it creates a small ritual that reinforces closure.

- **State 3 (Ready to Claim):** Timer hits 00:00. The screen should change colour to orange (or blink once orange), and show “COMPLETE” prominently. The Confirm button (orange, medium-size) lights up. The rest of UI dims or hides.
- **Confirmation:** User presses “CONFIRM”. Upon press, display a quick “+N pts” overlay (e.g. “+3”). Then return to Idle (State 1).
- **Failure mode:** If we auto-clear without confirm, the user might miss rewarding themselves. If we *don’t* allow confirm, the session ends abruptly which can feel abrupt. Confirm button adds one more intentional step.

Testable hypothesis: *With confirmation vs auto-complete*, does requiring confirm increase subjective satisfaction or completion self-report? We can measure whether users skip pressing confirm when distracted.

11. Checkpoint “Pings”

Optionally, the device can use the NFC system as *wake-up checkpoints*. For example, tapping a “home tag” could trigger a brief message: “HOME – continuing Maths (6:32 left)”. This would be a minimal overlay (e.g. 1 line, grey text) that fades after 1s. Similarly, a “coffee break” button might show a quick tip. The key is *brevity*.

We should avoid passive reminders like “You should be working!”. Instead, any prompt should be purely factual: “Remember, next is Maths 20:00” if certain conditions are met. After display, revert to normal state automatically.

12. “Make Laziness Easy”

This overarching principle means the path of least resistance accomplishes the task. Concretely:

- **Preselected task:** The Idle screen should already highlight the *next* task (TODAY → next task on top, under “NEXT:”). The dial can change tasks, but the default (first task) is preselected.
- **Preselected duration:** The screen could default to a standard duration (e.g. 20:00) when a task is chosen. User can tweak via dial if needed.
- **One-touch start:** With default task/time, the user only presses START to begin. No menus, no confirmations beyond that.
- **Resume session:** If a paused session exists, instead of showing tasks, the device could first offer “RESUME MATHS – 6:32 remaining” (see Adaptive Defaults). This is likely easier than restarting a new session.

Potential failure: The user may accidentally start an unintended task if the default is wrong. Always allow cancel or dial-turn first. Also, if the user persistently changes the default time, adapt (see Adaptive Defaults).

13. NFC Implementation Intentions

Examples of NFC-use scenarios:

- **Morning Tag:** When tapped first thing, display “GOOD MORNING – Today: 3 tasks (Maths, English, Chemistry)”. This orients the user. The tag action might also dim/snooze any night mode.
- **Arrival Tag:** When the user taps a “Desk/Home” tag, show “NEXT TASK – Chemistry: Atoms Q1–8 – 20:00” and a prominent START button. The idea: *cue* → *start homework*.
- **Task-complete Tag:** If the user scans a “Done tag” (like leaving the desk or scanning a sticker after finishing), the device could show points accumulated and move to the next task.
- **Anytime Task Change:** A general “HOMEWORK” tag might bring up the idle task list immediately.

The mermaid flowchart below illustrates a simplified logic: an NFC event leads to a specific display or action, then returns to normal idle.

flowchart TD

```
A[Idle Screen] -->|Scan Tag| B{Tag Type}
B -->|Morning/Home| C[Display Today's Tasks (Idle state)]
B -->|Desk/Start| D[Show Next Task and START]
B -->|Break Tag| E[Suggest Move 30 sec, show next task]
C --> A
D --> A
E --> A
```

Hypothesis: Test sessions with vs without NFC tags for context cues. Measure if NFC reminders increase session start-rate or decrease missed tasks.

14. Gamification / Points (Minimalist)

Our approach: **lightweight reinforcement**. Points (“Today: 24 pts”) can appear on the Idle screen in small text or LED indicator. Avoid large scoreboards.

- **Points per task:** E.g. +1 for start, +2 or +3 for completion. Show them in small orange text at bottom.
- **No streak/punishment:** Do **not** implement streak counters or penalty points. ADHD research suggests reward is more effective than punishment [31†L127-L135]. Also, streaks can demotivate if broken.
- **Descriptive summaries:** Periodically (e.g. NFC events) display “3 tasks done, 47 min total” instead of “GREAT JOB!”. Instrument-like readouts feel more neutral.
- **Testable metric:** Compare an interface with points vs identical one without (or with neutral feedback) and measure motivation/self-report. Possibly track whether points correlate with actual time-on-task.

15. Handling Interruptions and Pauses

A critical design is *state persistence*. When the user pauses or walks away, the device should:

- **Save the exact session state.** On PAUSE (manually or automatically if possible), show “PAUSED – 6:32 remaining – Press RESUME”.
- **No penalty:** Use neutral language (“PAUSED”) not “FAILED”. Return feeling should be: “We still remember you had 6:32 left.”
- **Automatic resume suggestion:** When the user comes back (screen reactivated or tap), the device could immediately show “RESUME [Task] – 6:32 remaining” instead of the full task list. A single START press should then continue.
- **Interrupt mode:** If they started a session and a major break (like an NFC home tag) happens, the device could auto-switch to a “Checkpoint overlay” (State 4) saying e.g. “HOME – [Task] 6:32 left” for 1s, then go to Idle.

This aligns with “make cancellation harder”: to cancel a session entirely, require extra gestures (hold+turn or confirm). Otherwise, assume intention persists.

Evidence: Prospective memory theory suggests that linking tasks to context (e.g. returning to workspace) improves recall [36†L426-L435]. Also, constant interruptions (like negative messaging) are known to frustrate ADHD users.

16. Micro-Breaks and Movement

Rather than enforcing Pomodoro breaks rigidly, we suggest *micro-break prompts*. For example, on completing a session:

- **Break screen:** Show a quick 30–60 sec “BREAK” screen with a suggestion (“Stretch your arms and walk a bit”). Possibly a simple animated icon (a person walking) at low contrast. The next task remains visible (“Next: Maths 12.3”).
- **Encourage movement:** Meta-analyses show exercise improves ADHD executive function [50†L285-L293]. Short bursts of movement (even standing up) can boost subsequent focus.
- **Time the break:** Make break length short (like 30s) and not mandatory to “complete”. After the time or upon user pressing “START” again, resume next session.

This uses the transition time to incorporate beneficial activity without derailing routine.

17. Adaptive Defaults

Finally, embed *learning*: the device can track user choices and adapt defaults. For instance:

- If the user always changes “MATHS” from 20→10 min, after a few days default to 10.
- If they often skip tasks, lower point thresholds or shorten session to reduce drop-off.
- If they consistently start without adjusting, the chosen defaults are likely fine.

This can be an experimental feature: log user overrides, then update “profile” for each task.

18. Summary Table: Countdown vs Count-Up vs Hybrid

Attribute	Countdown	Count-up	Hybrid
Psychological affordance	Conveys <i>deadline</i> , finite end-point 【16†L52-L60】 ; motivates “almost done”	Emphasises <i>progress made</i> ; avoids pressing end-point	Combines both cues (progress + remaining) for fuller context
Evidence strength (ADHD)	Some support (ADHD deficits in time estimation) 【16†L52-L60】 , though no RCT on format	No strong ADHD-specific evidence; logical (achievement focus)	Untested in ADHD; theoretically informative
Distraction risk	Medium: digits update each second (ticking) can draw gaze	Medium: similar ticking; no visible end timer might be soothing	Higher if showing two changing elements; risk of clutter
Glanceability	High: “14:32” is direct; instinctive	High: “05:28” quick to read but unclear remaining time	High: shows both, but may need glance to parse meaning
Implementation complexity	Low (simple decrement)	Low (simple increment)	Moderate (need sync two displays or indicator)
Test metric	Task completion rate; subjective stress; number of clock-checks	Task persistence; perceived time estimation; check frequency	Same metrics, comparing against single-mode conditions
Motivational alignment	Emphasises end; may spur some, annoy others (variable)	Emphasises “I did X”; can be encouraging if user likes progress tracking	Offers flexible framing; user could choose via UI setting
User control	User may feel “tick” pressure; less choice needed	Requires mental mapping to stop criteria	Could allow user to switch (adds control)

Attribute	Countdown	Count-up	Hybrid
Data feedback	Direct measure of “time to go”	Encourages noticing own effort (“already this far”)	All info available if needed
Recommended default	Yes – leverages known timing deficits [16†L52-L60]	No – use as alternate mode	If uncertain, offer as secondary view to test performance

19. Experimental Validation Plan

We propose a mixed **within-subjects design** with ~30 teenage participants diagnosed with ADHD (age ~13–18, balanced gender). Each participant uses a simplified version of the device in controlled tasks (e.g. homework-like exercises) for a week, experiencing randomized conditions on key features. Key hypotheses to test include:

1. **Countdown vs Count-Up (within-subject):** Each subject has two weeks: one with countdown and one with count-up (order counterbalanced). Measure *time-on-task* and *session completion rate*, as well as self-reported focus and stress.
2. **Progress Bar vs Numeric-only:** Compare sessions where a subtle progress bar is shown vs hidden. Metrics: *task persistence* and subjective usefulness rating.
3. **One-press Start (default task) vs Manual Select:** Days with preselected task vs days where a task-list must be navigated. Metric: *start latency* and number of cancelled starts.
4. **Immediate Points vs Delayed Score:** On some sessions, award +1 immediately at start vs no start bonus (only end reward). Measure motivation (e.g. frequency of start) and enjoyment.
5. **Subtle Pulse Feedback vs None:** Enable a short screen blink at session start/end on some days. Check if it affects *return latency* after distractions.
6. **NFC Cues vs None:** Provide environmental tags triggering reminders on some days. Track *missed sessions* and *self-report of remembering tasks*.
7. **Confirmation Required vs Auto-Complete:** Require confirm on some days vs auto-mark complete. Survey preference and count of unfinished sessions.
8. **Micro-break Prompts vs None:** On some breaks suggest movement vs just a timer. Evaluate subsequent session focus (quiz accuracy or on-task behaviour) and user perceived refreshment.

Participants would keep daily logs (qualitative) and device logs (quantitative: start/stop timestamps). Primary outcomes: *session start rate*, *time on task*, *return delay after pause*, and user ratings of burden/annoyance. Sample size 30 yields >80% power to detect medium effects ($d \approx 0.6$) at $\alpha=0.05$, within-subject. The protocol would be approved by ethics (parental consent and teen assent; data anonymised; right to withdraw).

Metrics: We will compare means (paired t-tests or mixed ANOVAs) and use regression for engagement factors. Questionnaire items (Likert) on perceived helpfulness of cues. Negative effects (e.g. increased procrastination) will also be documented.

20. UI Deliverables & Specifications

Front Panel Layout (to scale)

(All dimensions in mm)

- **Housing:** Matte off-white plastic with exposed screws and chamfered edges (Teenage Engineering aesthetic).
- **Size:** Approx. 250×150 mm (landscape rectangle).
- **Top strip (branding):** ~30 mm high band across width. Wordmark “*Task Engine*” centered, with small tagline below (e.g. “task engine · NFC · wake-word”) in ~6 px text. Top-right corner: tiny icons (7×7 px) for Wi-Fi/Sync and battery (using white-on-black for status).
- **Main display cutout:** Off-centre left, sized for 51.2×35.7 mm (240×320 px rotated to 320×240). It sits ~10 mm below top strip and ~15 mm from left edge, leaving space on right for controls.
- **Dial:** Located to the right of screen, roughly centre-vertically. Outer knob Ø38 mm, 25 mm tall. Under it (below), two small buttons (Ø12–15 mm) labelled “START” and “PAUSE”.
- **NFC pad:** To the right of dial, a recessed Ø30–40 mm circular pad (with NFC icon or label “SCAN” above). Next to pad: small “CONFIRM” button.
- **Bottom edge strip (~15 mm high):** Visual separation line. Contains two slide switches: “mic” (mute) and “power” (on/off), each ~13×6 mm. Labels small text (8px). Also a USB-C port recess (~9×3.4 mm) for charging, on left side.
- **Speaker:** A 50 mm round grille (dot-matrix style) somewhere left of dial (maybe below screen), with small LED cluster (~40×40 mm area) for status lights near it.
- **Dimensions summary:** Precise panel spacing (for prototype, front panel drawn to scale 250×150 mm). The screen active area (40.8×30.6 mm) equates to the cutout above.

Screen States (240×320px active)

1. Idle – Today’s Overview:

2. *Layout:* Top-left: “TODAY” label (white, 12px) and date (e.g. “04 SEP”, 10px). Left area: a vertical list of tasks (max 3–4), each row ~40px high: first line “NEXT” (orange, 12px) above next task name (“CHEMISTRY”, white, 24px), and subtext (“Atoms · Q1–8”, grey 14px). Beneath that, other tasks in white 20px.
3. *Right side:* Points summary in small grey (“24 pts”, 14px) at bottom. The primary button is “START” (orange, 18px text on dark background) centered under next task info.
4. *Behaviour:* Screen is mostly static. Next task entry may have a thin orange underline. Date and points update daily.

5. Example text:

TODAY 04 SEP

NEXT
CHEMISTRY
Atoms · Q1–8

MATHS
Exercise 12

ENGLISH
Paragraph 2

24 pts
[START]

6. Active Session:

7. *Layout:* Top-left: task name ("CHEMISTRY", white 24px) and subtask ("Atoms · Q1–8", grey 14px). Center: large countdown digits in orange (e.g. "14:32", white 120px or orange 120px). Below digits: progress bar (8px tall) filled orange proportionally. Bottom-left: points (" +1" awarded, grey 14px). Bottom-right: "PAUSE" in grey 18px (highlighted outline only, since active).

8. *Behaviour:* Only the countdown number (and bar) updates. Everything else is frozen. No other animations.

9. *Example:*

CHEMISTRY
Atoms · Q1–8

14:32



PAUSE
(points: +1)

10. Ready-to-Claim:

11. *Layout:* After timer hits 00:00. The digits turn orange (or flash once) and show "00:00" bold (120px). Below: a label "COMPLETE" (orange 18px). A large orange button "[CONFIRM]" (18px). Task name remains at top (white).

12. *Behaviour:* The display area briefly pulses (e.g. orange glow for 0.2s) to draw attention. After pulse, screen is static until user presses CONFIRM.

13. *Example:*

CHEMISTRY

00:00

COMPLETE

[CONFIRM]

14. NFC Checkpoint Overlay:

15. *Layout*: Very minimal. For example, after home-tag: one grey text line overlaid on screen: "HOME – MATHS 06:32 left" (grey 16px) on a black background. No orange. Display for ~1s then revert to Idle (State 1).

16. *Behaviour*: No sound; just fade in/out quickly.

17. Night/Dimmed Mode:

18. *Layout*: All pixels off or only a single very dim white dot/indicator. The panel appears "asleep."

19. *Behaviour*: No button interaction (except power). If an alarm or word is recognized at night, switch briefly to wake mode (State 4 style) then back off.

(Optional) Dial Close-up:

- **Diameter**: 38 mm, anodised sand-blast aluminium (suggest #6061-T6).
- **Marking**: A small engraved line indicator. Edge knurled or ridged for grip.
- **Position**: Finger accessible; protrudes ~2–3 mm beyond front panel.
- **UX note**: Each click turn should give a gentle tick (tactile/vibration). The device firmware should "latch" the display on task change with this tick.

Fonts & Colors Recap:

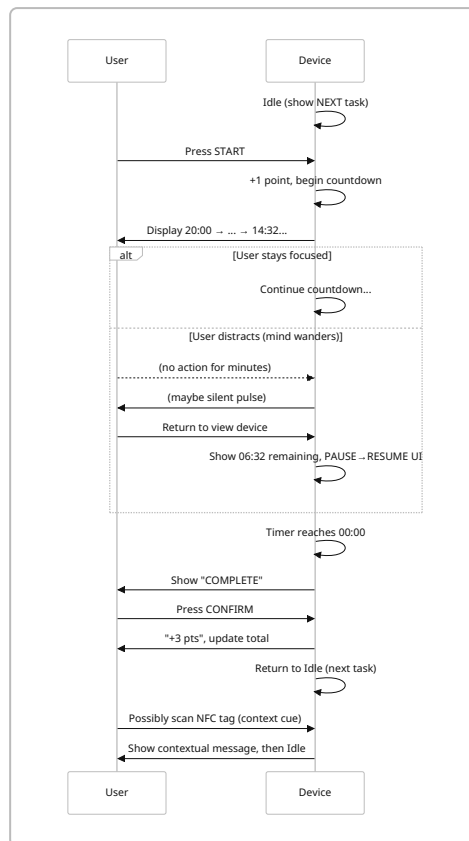
- **Primary accent (orange)**: #FF6633 (RGB 255,102,51).
- **Background**: #121212 (near-black).
- **Light text**: #FFFFFF; **grey text**: #CCCCCC or #888888 for less important info.
- **Font sizes (for 240×320 screen)**: 120px (timer), 24px (task name), 18px (buttons/labels), 14px (subtext/points), 10px (date/micro-labels), 8px (printed spec).
- **Progress bar**: Fill color same orange. Bar background: dark grey (#333333).

21. Mermaid Diagrams

- **State Machine (Dial Interaction)**:

```
stateDiagram-v2
    [*] --> Idle
    Idle --> SelectingTask : Turn Dial
    SelectingTask --> Idle : Stop Turning
    SelectingTask --> Active : Press START
    Active --> Paused : Press PAUSE
    Paused --> Active : Press RESUME
    Active --> ReadyToClaim : Timer=00:00
    ReadyToClaim --> Idle : Press CONFIRM
    Paused --> Idle : Hold+Press CANCEL
    Idle --> [*] : Power Off
```

- **Session Timeline (Sequence)**:



• NFC Flowchart:

flowchart TD

Start([Idle Screen])

NFC{NFC Event?}

NFC -->|Morning Tag| Morning[Show "Good Morning – 3 tasks"]

NFC -->|Desk Tag| Desk[Show NEXT task & START]

NFC -->|Home Tag| Home[Show "Welcome Back – NEXT: Maths"]

NFC -->|Other| None[No Change]

Morning --> Start

Desk --> Start

Home --> Start

None --> Start

(Legends: Orange items are user actions, blue are device states.)

Each recommendation above can be validated by targeted A/B testing or prototyping (e.g. an app simulating the screen) with ADHD teens. The combination of evidence-based mechanisms (timing aids [16†L52-L60] , frequent reinforcement [31†L127-L135] , cue-driven prompts [36†L426-L435]) aims to maximize focus support while avoiding additional cognitive load or distraction.

Sources: Peer-reviewed studies and meta-analyses on ADHD timing, reward, prospective memory and exercise [16†L52-L60] [31†L127-L135] [36†L426-L435] [41†L50-L59] [50†L285-L293] . Findings are also informed by HCI principles on minimal UIs and cognitive support. The above in-text citations link to key results in the literature.
